ICS Project 1

2024-04-15

```r
knitr::opts_chunk$set(echo = TRUE)
rm(list = ls())
setwd("C:/Users/rahul/OneDrive/Desktop/Notes/SS-24/ICS")
```

Data Processing

```r
# Required Libraries
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(corrplot)
```

```
## corrplot 0.92 loaded
```

```r
library(ggplot2)

data_set_original <- read.csv2(file = "Happiness.csv", header = TRUE)

# Loading data set and filter
data_set <- read.csv2(file = "Happiness.csv", header = TRUE)
names(data_set)[c(4,7,18)] <- c("Log.GDP", "Life.Freedom", "avg.gini")
data_set <- data_set[,c("Country.name", "Year", "Life.Ladder", "Log.GDP",
                        "Life.Freedom", "GINI.index..World.Bank.estimate.",
                        "avg.gini")]
# Data Insights
# head(data_set)
# names(data_set)
```

Task 1 ==========

```r
##================ a) Data set with Max Year =======================

## Data set with their last years data
data_last.year <- data_set %>%
  group_by(Country.name) %>%
  filter( Year == max(Year) )
```

```r
data_last.ladder.1 <- data_last.year[, c("Country.name", "Year",
                                         "Life.Ladder", "Log.GDP",
                                      "Life.Freedom")]

sum(is.na(data_last.ladder.1$Life.Ladder))
```

```
## [1] 0
```

```r
abs(5.472-5.461)
```

```
## [1] 0.011
```

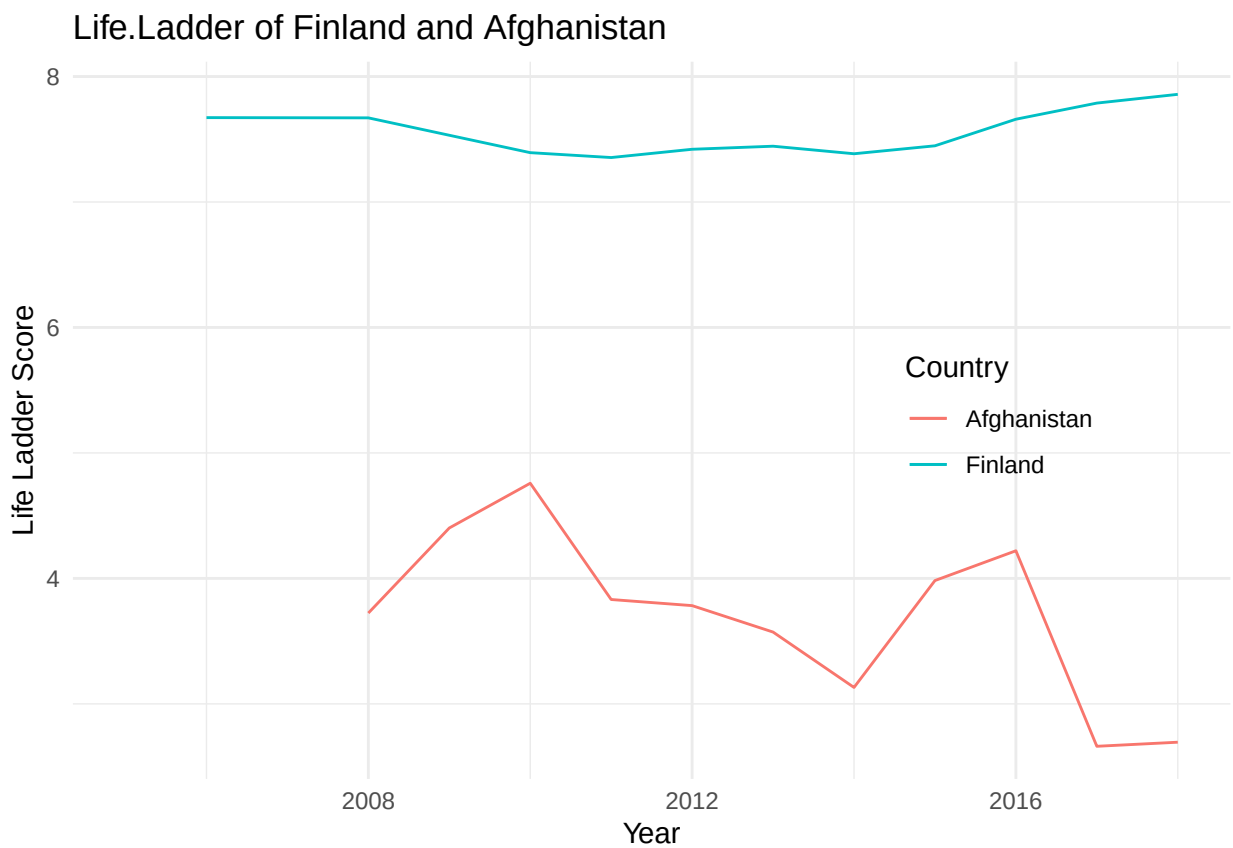
```r
# 3 Countries with maximum Life.Ladder in their last year
head(data_last.ladder.1[order(data_last.ladder.1$Life.Ladder,
                              decreasing = TRUE), ],n =3)
```

```
## # A tibble: 3 x 5
## # Groups:   Country.name [3]
##   Country.name  Year Life.Ladder Log.GDP Life.Freedom
##   <chr>        <int>       <dbl>   <dbl>        <dbl>
## 1 Finland       2018        7.86    10.6        0.938
## 2 Denmark       2018        7.65    10.8        0.935
## 3 Switzerland   2018        7.51    11.0        0.926
```

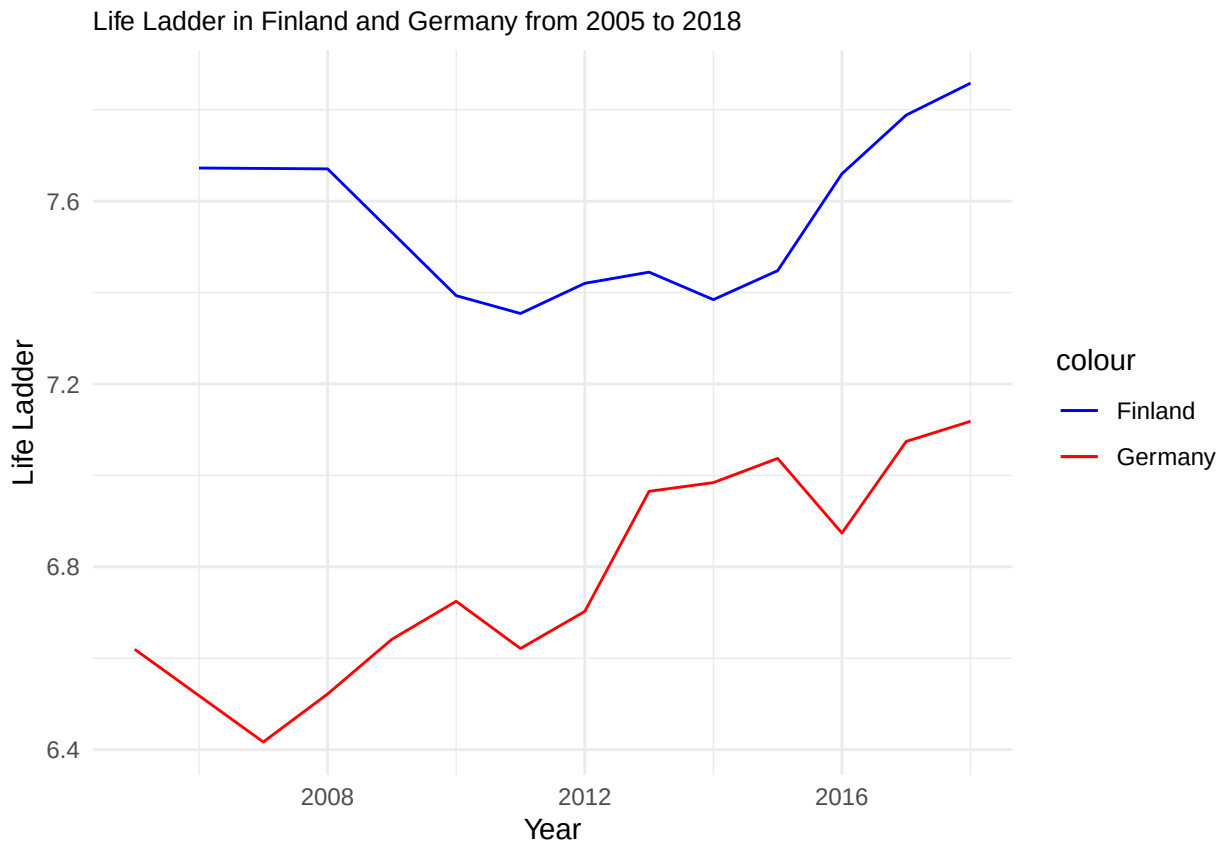
```r
# 3 Countries with minimum Life.Ladder in their last year
tail(data_last.ladder.1[order(data_last.ladder.1$Life.Ladder,
                              decreasing = TRUE), ],n =3)
```

```
## # A tibble: 3 x 5
## # Groups:   Country.name [3]
##   Country.name  Year Life.Ladder Log.GDP Life.Freedom
##   <chr>        <int>       <dbl>   <dbl>        <dbl>
## 1 Yemen         2018        3.06   NA           0.553
## 2 South Sudan   2017        2.82   NA           0.456
## 3 Afghanistan   2018        2.69    7.49        0.374
```

```r
data_finland <- data_set[which(data_set$Country.name == "Finland"), ]
data_afghanistan <- data_set[which(data_set$Country.name == "Afghanistan"),]
data_germany <- data_set[which(data_set$Country.name == "Germany"),]
data_india <- data_set[which(data_set$Country.name == "India"),]

## Graph of Ladder of both countries
ggplot() +
  geom_line(data = data_finland,
            aes(x = Year, y = Life.Ladder, col = "Finland")) +
  geom_line(data = data_afghanistan,
            aes(x = Year, y = Life.Ladder, col = "Afghanistan")) +
  labs(title = "Life.Ladder of Finland and Afghanistan",
       x = "Year",
       y = "Life Ladder Score", color = "Country") +
  xlim(2005, 2018) +
  theme_minimal() +
  theme(legend.position = c(0.8, 0.5))
```




```r
## Trend of Ladder in Finland and Germany
library(ggplot2)

ggplot() +
  geom_line(data = data_finland, aes(x = Year, y = Life.Ladder,
                                     col = "Finland")) +
  geom_line(data = data_germany, aes(x = Year, y = Life.Ladder,
                                     col = "Germany")) +

  labs(title = "Life Ladder in Finland and Germany from 2005 to 2018",
       x = "Year",
       y = "Life Ladder") +
  scale_color_manual(values = c(Finland = "blue", Germany = "red")) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))  # Adjust title size here
```




```
#No. of countries those do not have data for 2018
nrow(data_last.ladder.1[which(data_last.ladder.1$Year != 2018),]) #29
```

```
## [1] 29
```

```
data_not.in.2018 <- data_last.ladder.1[data_last.ladder.1$Year != 2018, ]
data_not.in.2017.18 <- data_not.in.2018[data_not.in.2018$Year != 2017, ]
nrow(data_not.in.2017.18) #13
```

```
## [1] 13
```

```
summary(data_last.ladder.1$Life.Ladder)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   2.694   4.659   5.472   5.461   6.235   7.858
```

```
#%%%
```

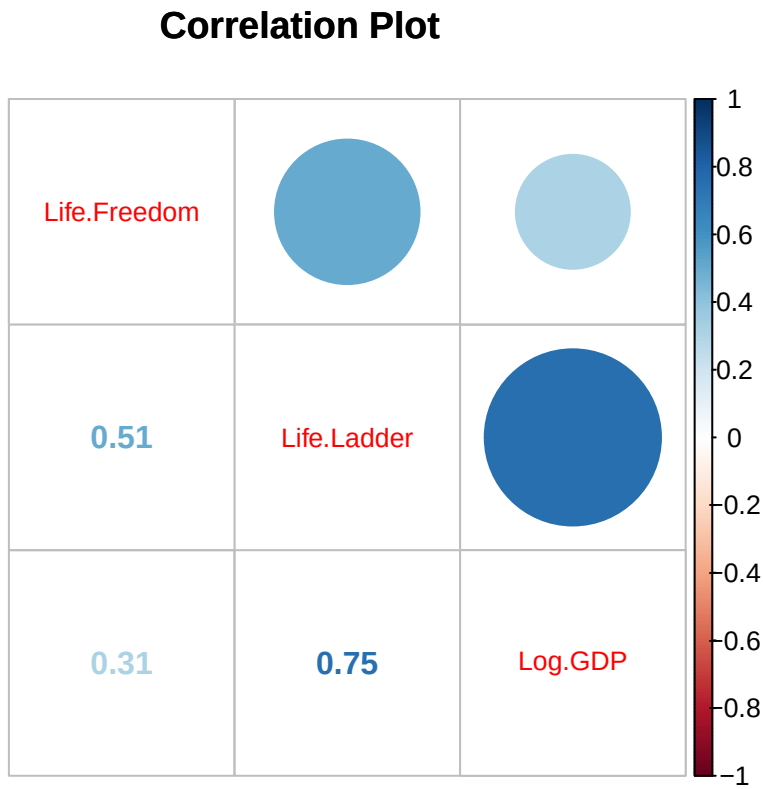
```r
# Scatter plot for Life.Ladder vs Log.GDP
g.1.1 <- ggplot(data = data_last.correlation.1, aes(x = Log.GDP, y = Life.Ladder)) +
  geom_point(color = "black", shape = 4) +
  labs(title = "Life Ladder vs Log GDP",
       x = "Log GDP",
       y = "Life Ladder Score") +
  theme_minimal()

# Scatter plot for Life.Ladder vs Life.Freedom
g.1.2 <- ggplot(data = data_last.correlation.1, aes(x = Life.Freedom, y = Life.Ladder)) +
  geom_point(color = "black", shape = 4) +
  labs(title = "Life Ladder vs Life Freedom",
       x = "Life Freedom",
       y = "Life Ladder Score") +
  theme_minimal()


(corr.matrix <- cor(data_last.correlation.1[, c("Life.Ladder", "Log.GDP",
                                                "Life.Freedom")]))
```

```
##              Life.Ladder   Log.GDP Life.Freedom
## Life.Ladder    1.0000000 0.7530505    0.5064871
## Log.GDP        0.7530505 1.0000000    0.3147273
## Life.Freedom   0.5064871 0.3147273    1.0000000
```

```r
corr.matrix
```

```
##              Life.Ladder   Log.GDP Life.Freedom
## Life.Ladder    1.0000000 0.7530505    0.5064871
## Log.GDP        0.7530505 1.0000000    0.3147273
## Life.Freedom   0.5064871 0.3147273    1.0000000
```

```r
corrplot.mixed(corr = corr.matrix, order = 'AOE', main = "Correlation Plot",
               mar=c(1, 1, 3, 2), tl.cex = 0.8)
```



Task 2 ==========

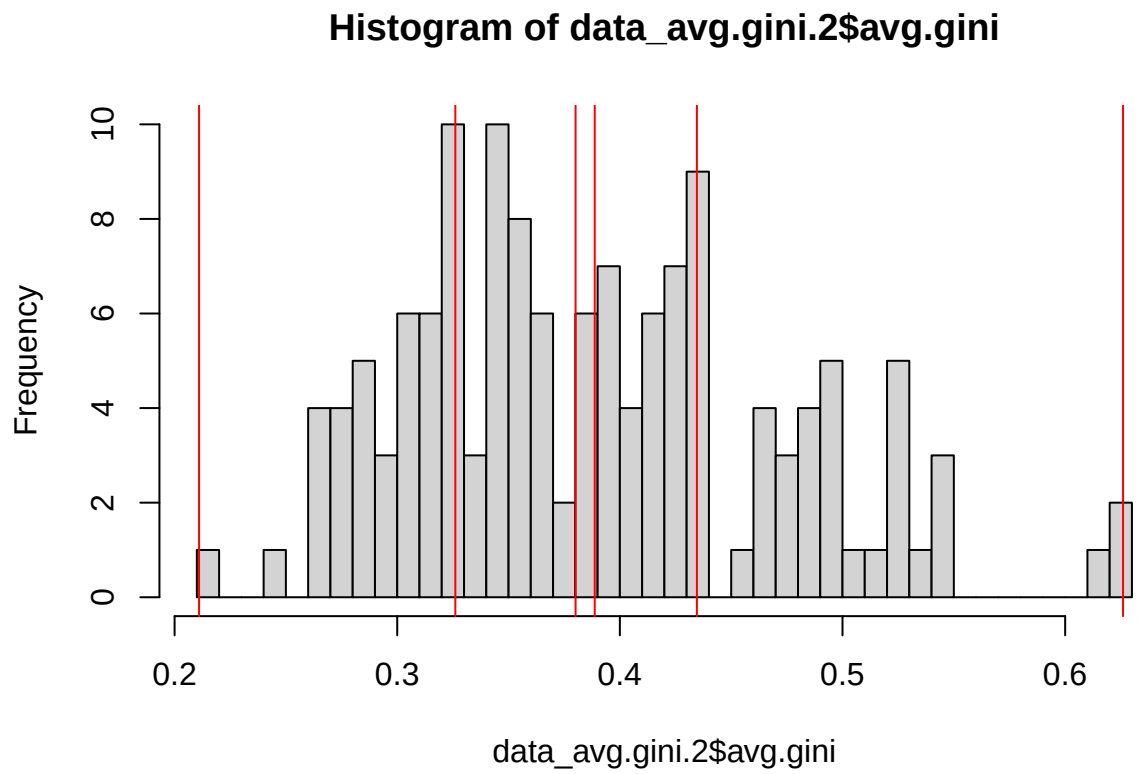
```
colSums( is.na(data_set[,c("GINI.index..World.Bank.estimate.",
                           "avg.gini")]) )
```

```
## GINI.index..World.Bank.estimate.                         avg.gini
##                             1061                              202
```

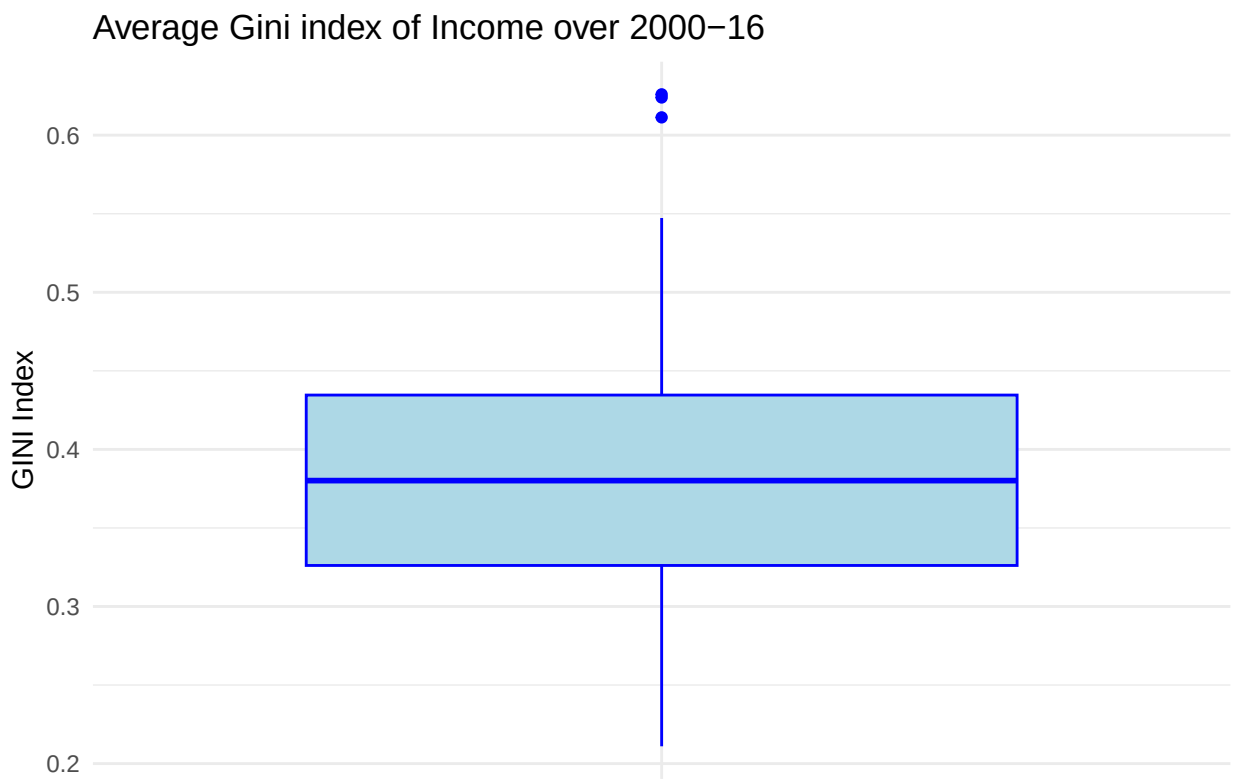
```
data_avg.gini.2 <- na.omit( data_set[ , c("Country.name", "Year", "avg.gini")] )

data_avg.gini.2 <- data_avg.gini.2 %>%
  group_by(Country.name) %>%
  filter( Year == max(Year) )

summary(data_avg.gini.2$avg.gini)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.2110  0.3261  0.3801  0.3887  0.4346  0.6260
```

```
hist(data_avg.gini.2$avg.gini, breaks = 30)
abline( v = c(0.2110, 0.3261, 0.3801, 0.3887, 0.4346, 0.6260), col = "red")
```




```r
# Box Plot of Gini INdex
ggplot(data_avg.gini.2, aes(x = "", y = avg.gini)) +
  geom_boxplot(fill = "lightblue", color = "blue") +
  labs(title = "Average Gini index of Income over 2000-16" ,
       y = "GINI Index", x ="") +
  theme_minimal()
```




```r
head(data_avg.gini.2[order(data_avg.gini.2$avg.gini), ], n = 10)
```

```
## # A tibble: 10 x 3
## # Groups:   Country.name [10]
##    Country.name    Year avg.gini
##    <chr>          <int>    <dbl>
##  1 Azerbaijan      2018    0.211
##  2 Slovenia        2018    0.250
##  3 Czech Republic  2018    0.264
##  4 Ukraine         2018    0.265
##  5 Denmark         2018    0.267
##  6 Slovakia        2018    0.267
##  7 Norway          2018    0.272
##  8 Sweden          2018    0.274
##  9 Finland         2018    0.276
## 10 Algeria         2018    0.276
```

```r
tail(data_avg.gini.2[order(data_avg.gini.2$avg.gini), ], n = 10)
```

```
## # A tibble: 10 x 3
## # Groups:   Country.name [10]
##    Country.name  Year avg.gini
##    <chr>        <int>    <dbl>
##  1 Guatemala     2018    0.524
##  2 Zambia        2018    0.527
##  3 Lesotho       2017    0.529
##  4 Panama        2018    0.533
```

```
##  5 Colombia      2018    0.542
##  6 Honduras      2018    0.547
##  7 Brazil        2018    0.547
##  8 Namibia       2018    0.611
##  9 South Africa  2018    0.624
## 10 Botswana      2018    0.626
```

Task 3 ==========

```r
#removing GDF of World Bank Estimtes
data_set <- data_set[,c("Country.name","Year", "Life.Ladder", "Log.GDP",
                        "Life.Freedom", "avg.gini")]
summary(data_set)
```

```
##  Country.name            Year       Life.Ladder       Log.GDP
##  Length:1704        Min.   :2005   Min.   :2.662   Min.   : 6.457
##  Class :character   1st Qu.:2009   1st Qu.:4.611   1st Qu.: 8.304
##  Mode  :character   Median :2012   Median :5.340   Median : 9.406
##                     Mean   :2012   Mean   :5.437   Mean   : 9.222
##                     3rd Qu.:2015   3rd Qu.:6.274   3rd Qu.:10.193
##                     Max.   :2018   Max.   :8.019   Max.   :11.770
##                                                    NA's   :28
##   Life.Freedom       avg.gini
##  Min.   :0.2575   Min.   :0.2110
##  1st Qu.:0.6384   1st Qu.:0.3214
##  Median :0.7527   Median :0.3710
##  Mean   :0.7338   Mean   :0.3854
##  3rd Qu.:0.8482   3rd Qu.:0.4322
##  Max.   :0.9852   Max.   :0.6260
##  NA's   :29       NA's   :202
```

```r
summary(data_avg.gini.2) # 139 ie. 26 countries are missing
```

```
##  Country.name            Year         avg.gini
##  Length:139         Min.   :2011   Min.   :0.2110
##  Class :character   1st Qu.:2018   1st Qu.:0.3261
##  Mode  :character   Median :2018   Median :0.3801
##                     Mean   :2018   Mean   :0.3887
##                     3rd Qu.:2018   3rd Qu.:0.4346
##                     Max.   :2018   Max.   :0.6260
```

```r
summary(na.omit(data_last.ladder.1))
```

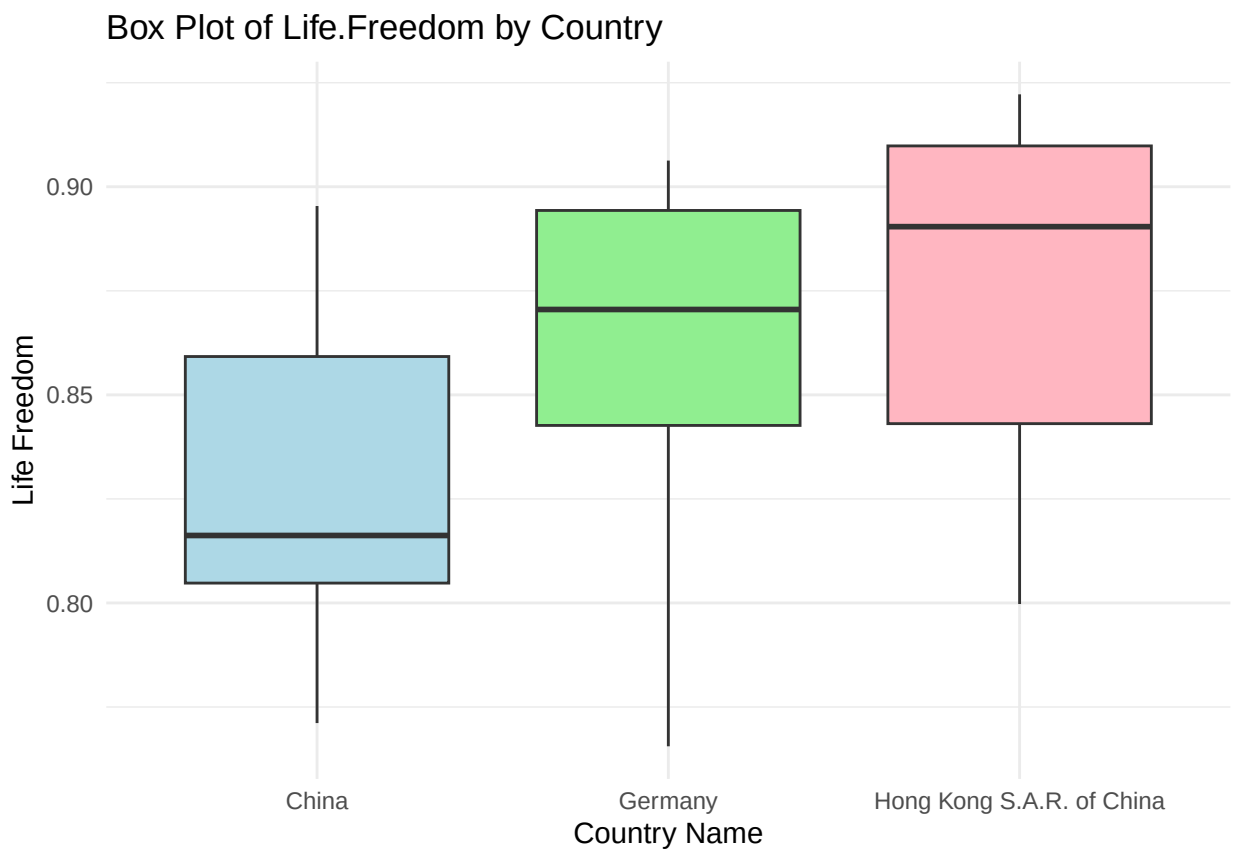
```
##  Country.name            Year       Life.Ladder       Log.GDP
##  Length:151         Min.   :2006   Min.   :2.694   Min.   : 6.494
##  Class :character   1st Qu.:2018   1st Qu.:4.661   1st Qu.: 8.345
##  Mode  :character   Median :2018   Median :5.465   Median : 9.412
##                     Mean   :2018   Mean   :5.481   Mean   : 9.242
##                     3rd Qu.:2018   3rd Qu.:6.203   3rd Qu.:10.203
##                     Max.   :2018   Max.   :7.858   Max.   :11.454
##   Life.Freedom
##  Min.   :0.2815
##  1st Qu.:0.7129
##  Median :0.7989
##  Mean   :0.7791
```

```
##  3rd Qu.:0.8765
##  Max.   :0.9699
```

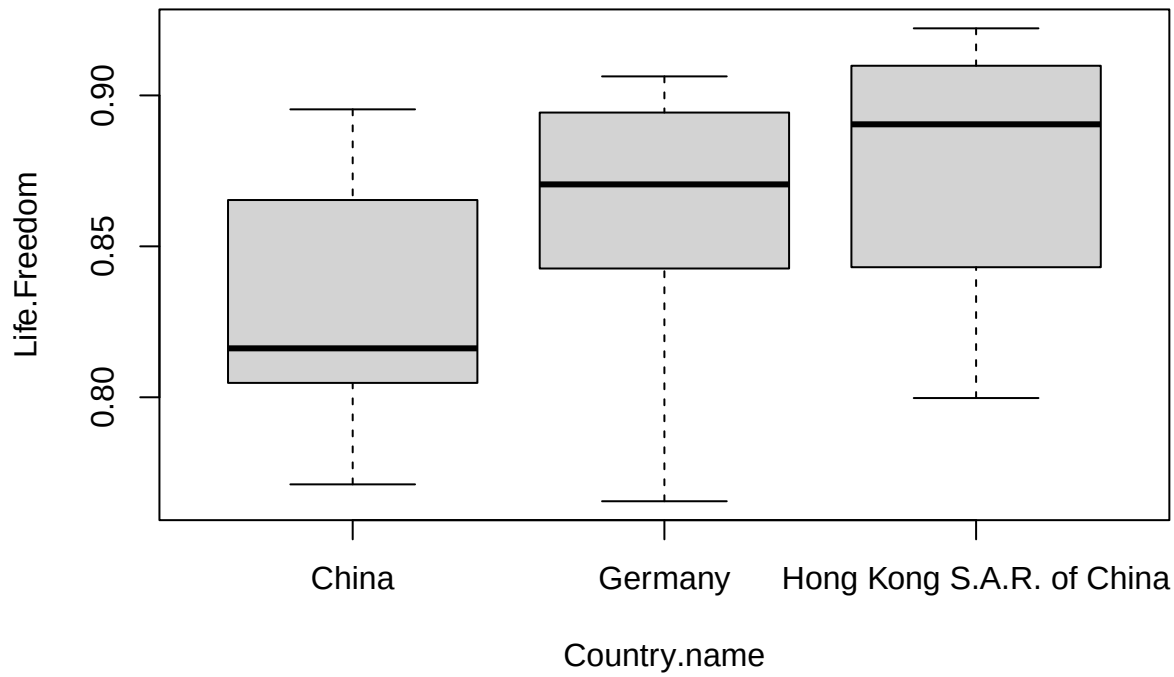
```r
## Sensitive questions to some countries
## Maybe, too personal to some culture
## Pessimistic about country
```

Task 4 ==========

```r
## Data Set for food comparison
data_set.food.4 <- data_set[which(data_set$Country.name == "Germany" |
                   data_set$Country.name == "Hong Kong S.A.R. of China" |
                   data_set$Country.name == "China"), ]
data_set.food.4 <- data_set.food.4[ , c("Country.name", "Year",
                                        "Log.GDP",
                                        "Life.Freedom")]
data_set.food.4 <- na.omit(data_set.food.4)

# Freedom to make choices -----------------------
ggplot(data_set.food.4, aes(x = Country.name, y = Life.Freedom,
                            fill = Country.name)) +
  geom_boxplot() +
  labs(title = "Box Plot of Life.Freedom by Country",
       x = "Country Name",
       y = "Life Freedom") +
  scale_fill_manual(values = c("lightblue", "lightgreen",
                               "lightpink", "lightyellow")) +
  theme_minimal() +
  theme(legend.position = "none")
```




```
summary(boxplot(Life.Freedom ~ Country.name,
        data = data_set.food.4)$stats)
```




```
##        V1               V2               V3
##  Min.   :0.7711   Min.   :0.7656   Min.   :0.7997
##  1st Qu.:0.8048   1st Qu.:0.8427   1st Qu.:0.8431
##  Median :0.8162   Median :0.8705   Median :0.8904
##  Mean   :0.8306   Mean   :0.8559   Mean   :0.8731
##  3rd Qu.:0.8653   3rd Qu.:0.8943   3rd Qu.:0.9098
##  Max.   :0.8954   Max.   :0.9063   Max.   :0.9222
```

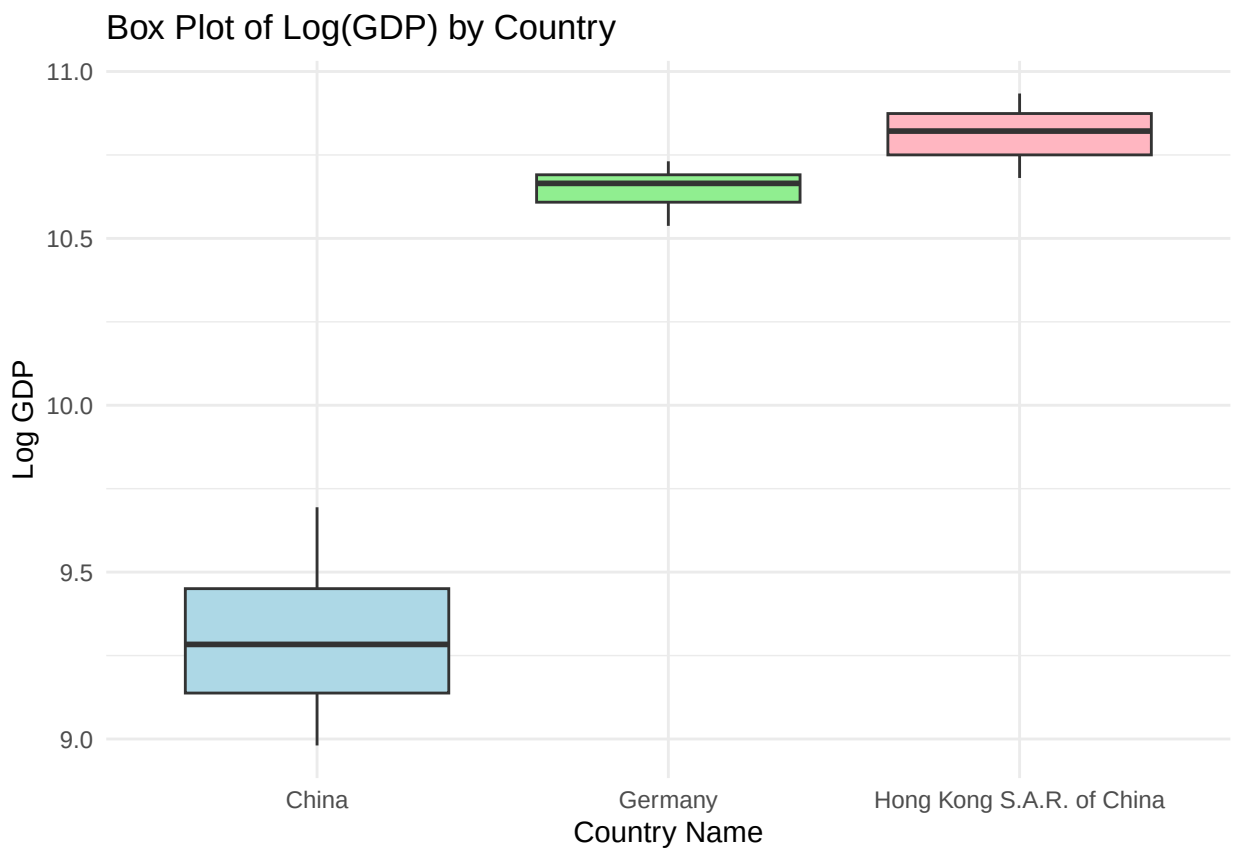
```r
#range
0.8954-0.7711
```

```
## [1] 0.1243
```

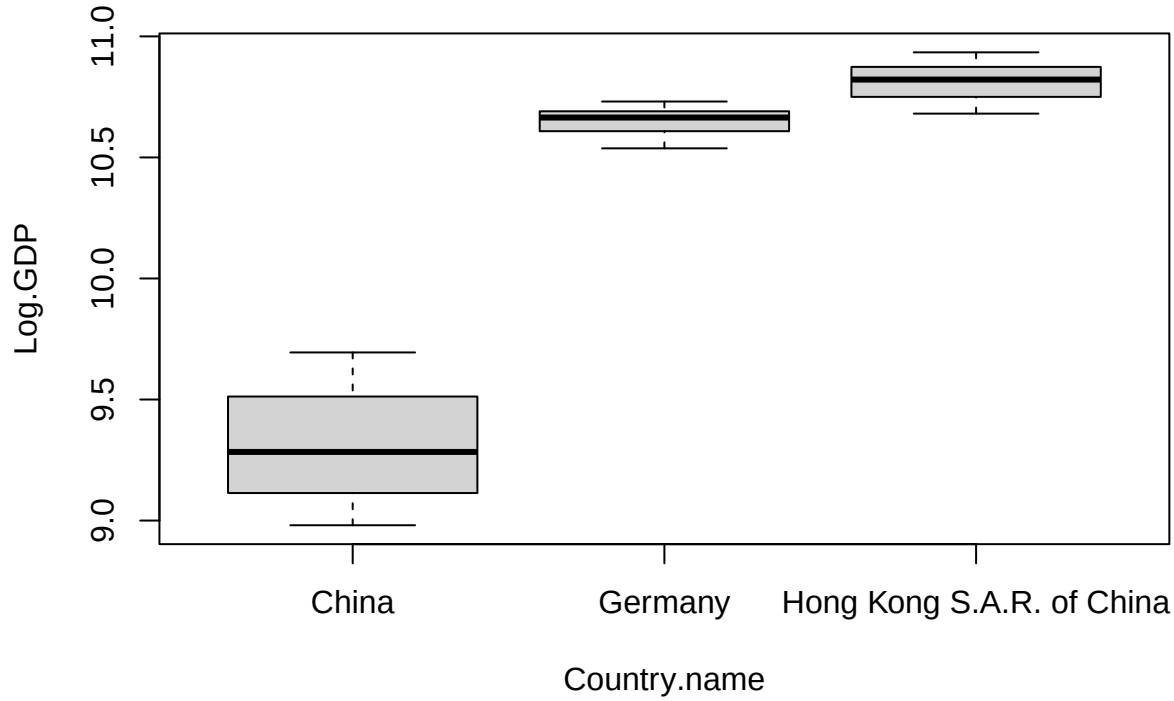
```r
# Hong Kong from Germany
0.8904  - 0.8705  #0.0199
```

```
## [1] 0.0199
```

```r
# Log.GDP ----------------------------
ggplot(data_set.food.4, aes(x = Country.name, y = Log.GDP,
                            fill = Country.name)) +
  geom_boxplot() +
  labs(title = "Box Plot of Log(GDP) by Country",
       x = "Country Name",
       y = "Log GDP") +
  scale_fill_manual(values = c("lightblue", "lightgreen",
                               "lightpink", "lightyellow")) +
  theme_minimal() +
  theme(legend.position = "none")
```




```
(summary(boxplot(Log.GDP ~ Country.name,
        data = data_set.food.4)$stats))
```




```
##        V1              V2              V3
##  Min.   :8.981   Min.   :10.54   Min.   :10.68
##  1st Qu.:9.114   1st Qu.:10.61   1st Qu.:10.75
##  Median :9.283   Median :10.66   Median :10.82
##  Mean   :9.317   Mean   :10.65   Mean   :10.81
##  3rd Qu.:9.512   3rd Qu.:10.69   3rd Qu.:10.87
##  Max.   :9.694   Max.   :10.73   Max.   :10.93
```

```r
# GDP differece of Hongkong - Germany
# 10.82 - 10.66 = 0.16
```

Task 5 ==========

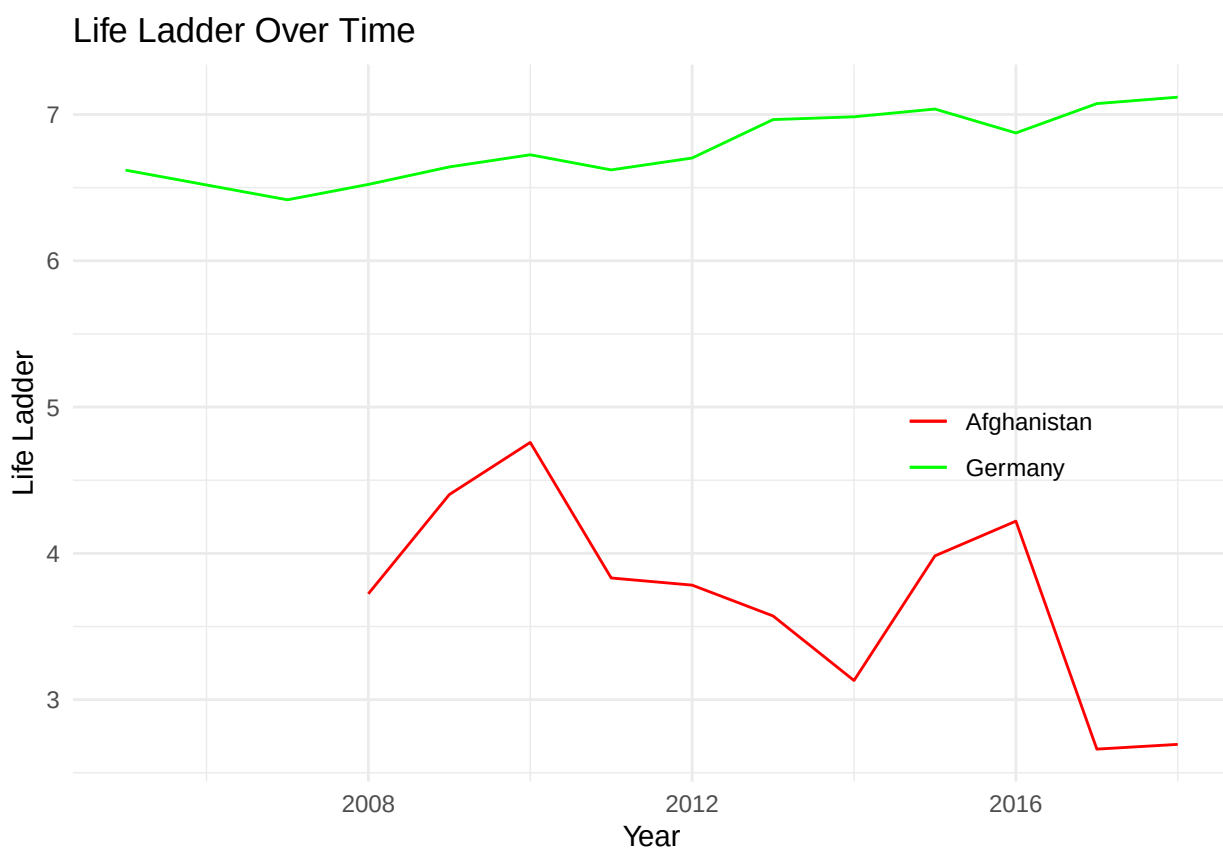
```r
## Measure of Central Tendency
## Afghanistan and Germany

data_afghanistan <- data_set[which(data_set$Country.name == "Afghanistan"),]
data_germany <- data_set[which(data_set$Country.name == "Germany"),]
data_india <- data_set[which(data_set$Country.name == "India"),]

(summary(data_afghanistan$Life.Ladder))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   2.662   3.351   3.783   3.705   4.102   4.758
```

```r
(summary(data_germany$Life.Ladder))
```

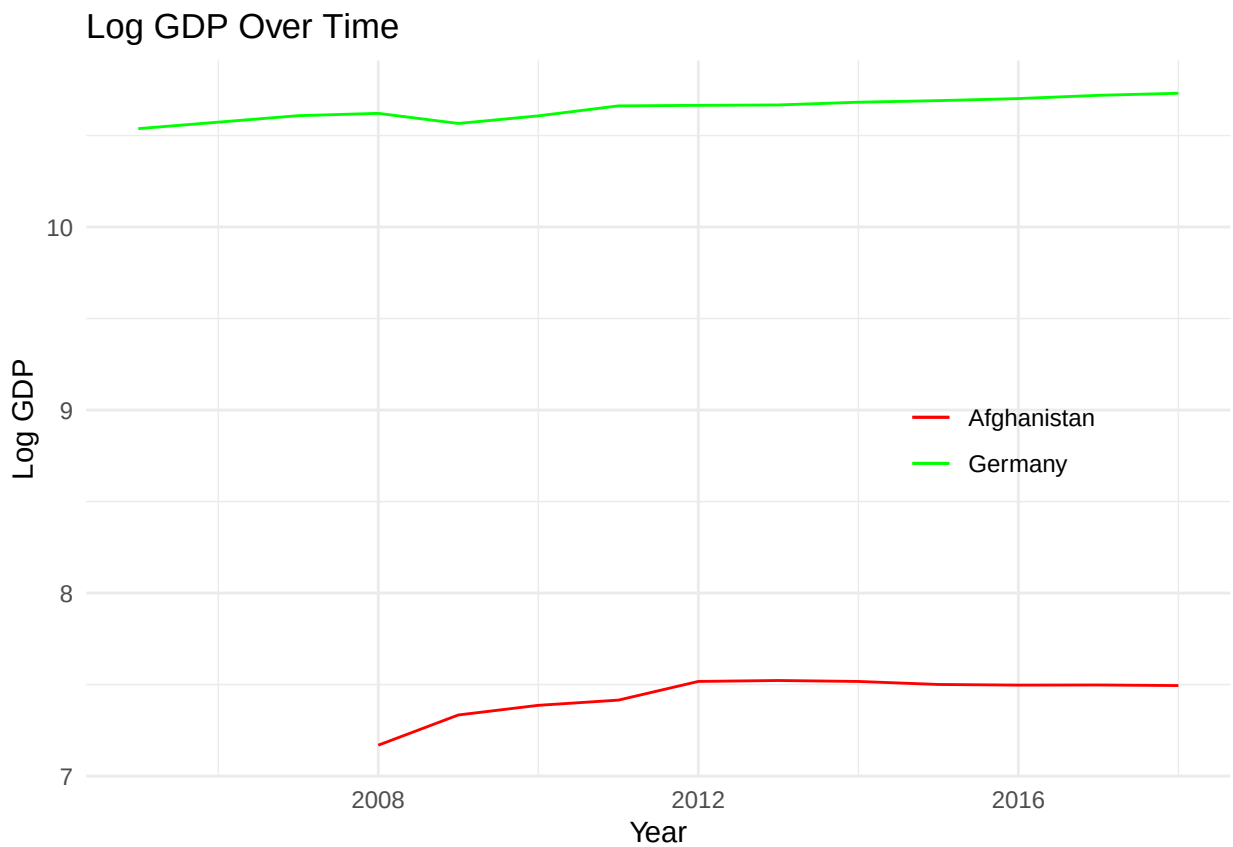
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   6.417   6.621   6.725   6.792   6.984   7.118
```

```r
# Life Ladder -----------------------------

ggplot() +
  geom_line(data = data_germany, aes(x = Year,
                                     y = Life.Ladder,
                                     color = "Germany")) +
  geom_line(data = data_afghanistan, aes(x = Year,
                                         y = Life.Ladder,
                                         color = "Afghanistan")) +
  labs(title = "Life Ladder Over Time",
       x = "Year",
       y = "Life Ladder",
       color = "") +
  theme_minimal() +
  scale_color_manual(values = c("Germany" = "green",
                                "Afghanistan" = "red")) +
  theme(legend.position = c(0.8, 0.5))
```



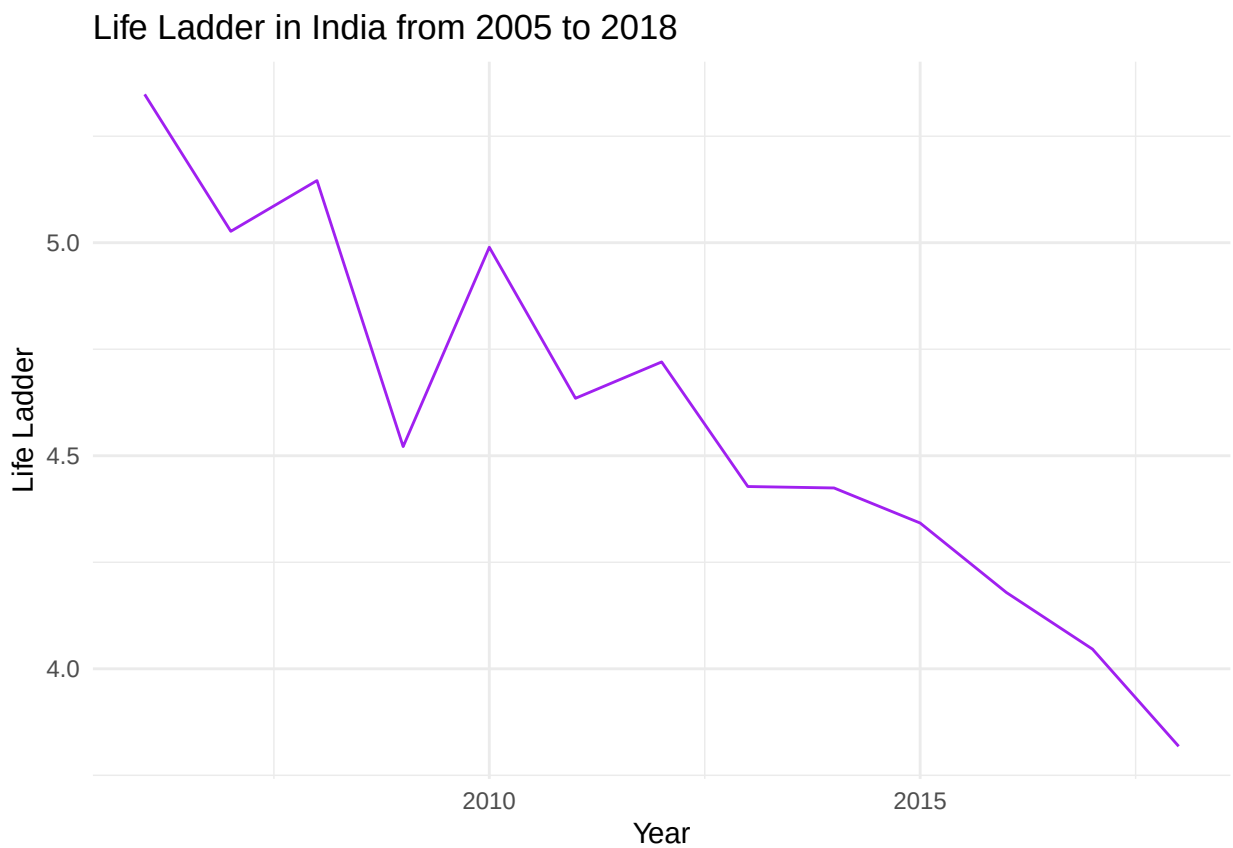

```r
# Log GDP -----------------------------
ggplot() +
  geom_line(data = data_germany, aes(x = Year,
                                     y = Log.GDP,
                                     color = "Germany")) +
  geom_line(data = data_afghanistan, aes(x = Year,
```

```r
                                        y = Log.GDP,
                                        color = "Afghanistan")) +
  labs(title = "Log GDP Over Time",
       x = "Year",
       y = "Log GDP",
       color = "") +
  theme_minimal() +
  scale_color_manual(values = c("Germany" = "green",
                                "Afghanistan" = "red")) +
  theme(legend.position = c(0.8, 0.5))
```

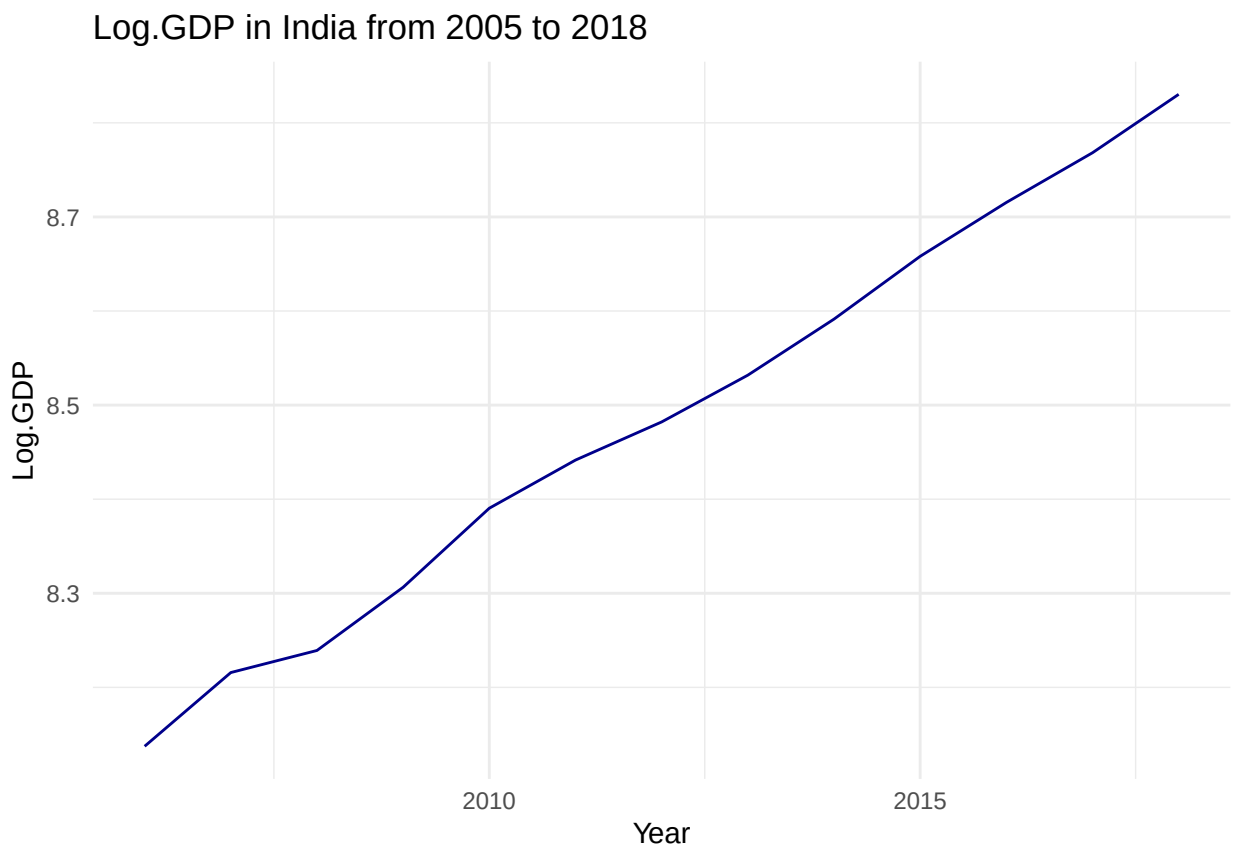


3rd Aspect

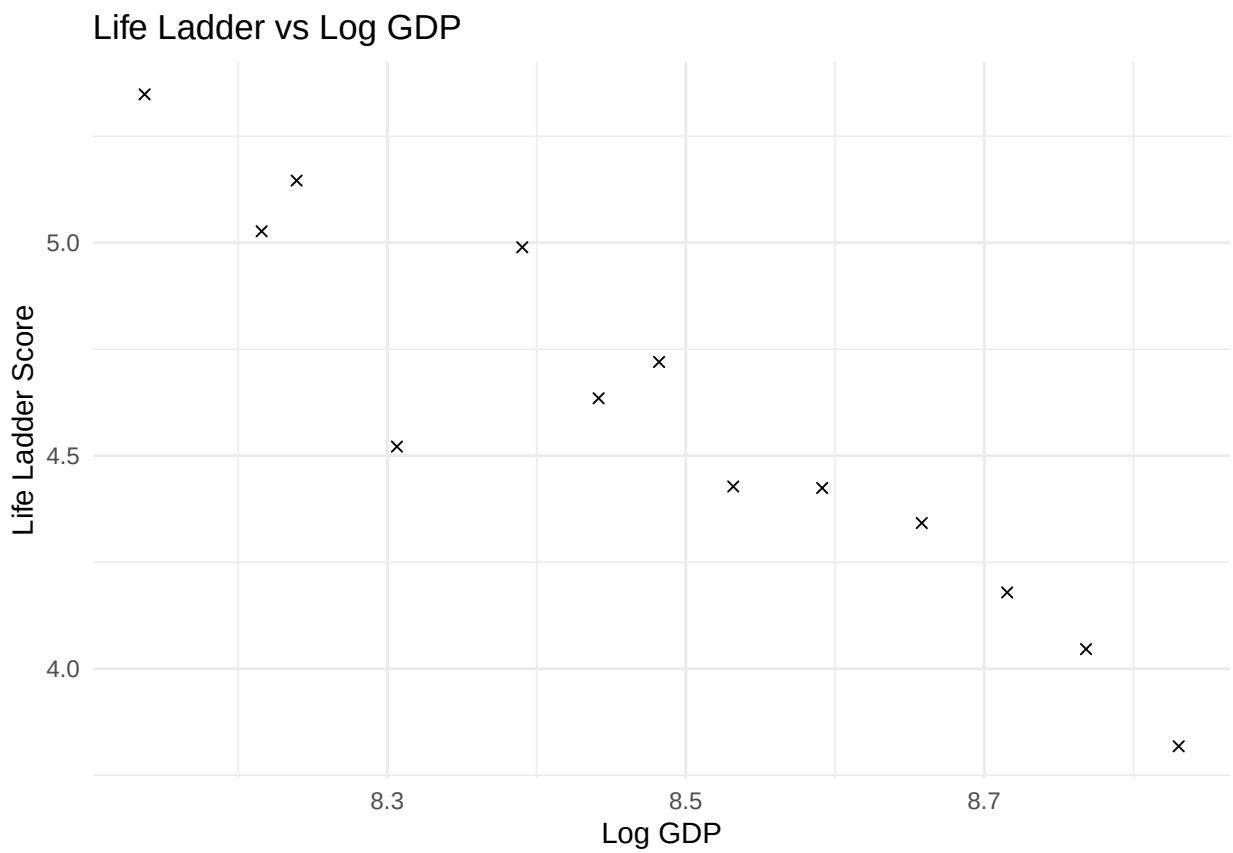
```r
# Life.Ladder and Log.GDP in India
ggplot() +
  geom_line(data = data_india, aes(x = Year,
                                   y = Life.Ladder), col = "purple") +
  labs(title = "Life Ladder in India from 2005 to 2018",
       x = "Year",
       y = "Life Ladder") +
  theme_minimal()
```




```r
ggplot() +
  geom_line(data = data_india, aes(x = Year,
                                   y = Log.GDP), col = "#00008B") +
  labs(title = "Log.GDP in India from 2005 to 2018",
       x = "Year",
       y = "Log.GDP") +
  theme_minimal()
```




```r
# Scatter plot for Life.Ladder vs Log.GDP
ggplot(data = data_india, aes(x = Log.GDP, y = Life.Ladder)) +
  geom_point(color = "black", shape = 4) +
  labs(title = "Life Ladder vs Log GDP",
       x = "Log GDP",
       y = "Life Ladder Score") +
  theme_minimal()
```


Life Ladder vs Log GDP
Life Ladder Score
5.0
4.5
4.0
8.3
8.5
8.7
Log GDP